

MLOps — Comprehensive Midterm Exam

PART A: MCQs (with Explanations)

Q1. In MLOps, what is the main reason models fail to reach production?

- A. Lack of model accuracy
- B. Poor user interface design
- C. Missing automation and deployment integration
- D. Incompatible data formats

Answer: C → Manual hand-offs between data science and ops break continuity. MLOps enforces CI/CD and automation .

Other options: A is short-term; B and D are peripheral issues.

Q2. In a feature store, *train-serve skew* occurs when —

- A. Features drift due to new data pipelines
- B. Live features differ in computation from training features
- C. Labels are missing in production
- D. Model overfits training data

Answer: B → Feature definitions differ between offline (train) and online (serve), causing degraded predictions .

Q3. Which command correctly exports a Conda environment to a portable YAML file?

- A. `conda export env.yaml`
- B. `conda env export > environment.yaml`
- C. `conda save env.yaml`
- D. `conda list --export`

Answer: B → This captures all dependencies and channels for reproducibility .

Q4. Containers differ from VMs because they —

- A. Require a hypervisor
- B. Share the host OS kernel
- C. Run slower than VMs
- D. Need separate CPU allocation per app

Answer: B → Containers share the kernel of the host and are lighter and faster .
A applies to VMs; C is reverse; D refers to VM behavior.

Q5. Which Dockerfile instruction sets the base image?

- A. RUN
- B. COPY
- C. FROM
- D. ENTRYPOINT

Answer: C → FROM centos:8 defines the base layer image .

Q6. A FastAPI app in a Docker container typically exposes port —

- A. 3000
- B. 8000
- C. 22
- D. 5000

Answer: B → Default port for Uvicorn/FastAPI is 8000.

Q7. Which Kubernetes scaling method adds more pods instead of increasing pod resources?

- A. Vertical Scaling
- B. Horizontal Scaling
- C. Diagonal Scaling
- D. Dynamic Threading

Answer: B → Horizontal Scaling creates replicas of pods to distribute load .

Q8. Which AutoML tool is open source and developed by Microsoft?

- A. Ludwig
- B. FLAML
- C. Vertex AI AutoML
- D. SageMaker Autopilot

Answer: B → FLAML (Fast Lightweight AutoML) is a Microsoft research project .

Q9. Which cloud service provides Feature Store and Model Monitor in one ecosystem?

- A. Azure AI Studio
- B. AWS SageMaker
- C. Google Vertex AI
- D. IBM Watson ML

Answer: B → SageMaker includes Feature Store, Model Monitor, Experiments .

Q10. The MLflow function to start tracking a run is —

- A. `mlflow.start_run()`
- B. `mlflow.begin()`
- C. `mlflow.run_experiment()`
- D. `mlflow.track()`

Answer: A → `mlflow.start_run()` begins a logging context for parameters and metrics.

Q11. Concept drift occurs when —

- A. The target distribution changes
- B. Input feature values change
- C. Input–output relationship changes
- D. Model overfits

Answer: C → Concept drift = change in relation between X and Y; data drift = change in X distribution .

Q12. CI/CD tools like GitHub Actions help MLOps by —

- A. Automating code testing and deployment
- B. Improving model accuracy
- C. Running data preprocessing locally
- D. Hosting ML models on prem

Answer: A → They integrate testing, Docker builds, and deployments for each commit .

Q13. Which command builds a Docker image named `mlapi:v1` from a Dockerfile?

- A. `docker create mlapi:v1`
- B. `docker build -t mlapi:v1 .`
- C. `docker run mlapi:v1`
- D. `docker push mlapi:v1`

Answer: B → `docker build -t name:tag .` builds the image from the Dockerfile.

Q14. Which metric is used to quantify data drift between train and live datasets?

- A. AUC
- B. PSI (Population Stability Index)
- C. RMSE
- D. F1-score

Answer: B → PSI measures distribution shifts across time.

Q15. In MLflow, the `tracking URI` defines —

- A. Location of data files
- B. Endpoint for model deployment
- C. Server where runs and metrics are stored
- D. GitHub repository link

Answer: C → Tracking URI points to local or remote MLflow server storing experiment data.

Q16. FastAPI is preferred over Flask in MLOps pipelines because —

- A. It is easier to learn
- B. It supports async requests and auto docs
- C. It is older and stable
- D. It needs no Dockerization

Answer: B → FastAPI is ASGI-based and auto-generates interactive docs (`/docs`).

Q17. The difference between horizontal and vertical scaling is that —

- A. Vertical adds pods, horizontal adds CPU
- B. Horizontal adds pods, vertical adds resources to pods
- C. Both add replicas
- D. Both use load balancers only

Answer: B → Horizontal = replication; Vertical = increase RAM/CPU on same node .

Q18. The Dockerfile command to define container entry execution is —

- A. RUN
- B. CMD
- C. ENTRYPOINT
- D. WORKDIR

Answer: C → **ENTRYPOINT** defines the main command executed when container runs .

Q19. In the MLOps lifecycle, the “model registry” stage is responsible for —

- A. Storing raw training data
- B. Storing and versioning approved models
- C. Running unit tests
- D. Generating synthetic data

Answer: B → Registry acts as a central hub for approved model versions before deployment .

Q20. Which service on GCP allows auto-scaling containerized APIs for ML models?

- A. GCE
- B. Cloud Run
- C. Cloud Storage
- D. BigQuery

Answer: B → Cloud Run hosts Docker containers serverlessly and auto-scales .



PART B: Short-Answer Questions (2–3 sentences each)

1 Explain the role of CI/CD in MLOps.

CI/CD automates model testing and deployment. Continuous Integration runs unit tests and quality checks, while Continuous Deployment builds containers and publishes new models to production safely without manual steps .

2 Differentiate between data drift and concept drift with examples.

Data drift means input features change (e.g., income distribution shifts); concept drift means their relationship to the output changes (e.g., spam words evolve). Both cause model accuracy drop over time.

3 How does a Feature Store help prevent train-serve skew?

It keeps a single definition of features for both training and serving. Thus, the same logic and data pipelines are reused in production, ensuring consistency .

4 Describe the difference between virtual environments and Conda environments.

Virtualenv is Python-only and uses pip, while Conda handles both Python and non-Python dependencies (C libs, CUDA). Conda is more robust for data science and MLOps reproducibility .

5 What is the purpose of a Docker image layer system?

Each Docker instruction creates a layer cached by the engine. This makes rebuilds faster and saves space because unchanged layers are reused between images .

6 Explain horizontal vs vertical scaling in Kubernetes with an example.

Horizontal scaling adds more pods for load distribution (e.g., 5 replicas of a FastAPI service). Vertical scaling adds CPU or RAM to the existing pod to handle larger requests .

7 What does the command `conda env update -f environment.yml --prune` do?

It updates the existing Conda environment to match the YAML file and removes packages not listed there. This keeps the environment clean and aligned with project dependencies .

8 How does MLflow support experiment tracking?

MLflow records parameters, metrics, and artifacts for each training run using a tracking URI. It helps compare experiments and retrieve the best performing model later.

9) Why are containers preferred over VMs in modern MLOps systems?

They start faster, use fewer resources, and run identically across platforms since they share the host kernel. VMs emulate hardware, making them heavier and slower .

10) Explain how cloud ML services like SageMaker and Vertex AI support end-to-end MLOps.

They cover data preparation, training, model registry, deployment, and monitoring under one platform. SageMaker has Experiments and Model Monitor, while Vertex AI integrates training, endpoints, and drift detection .